



Perioperative Management of Patients Receiving Anticoagulants

Introducing the Need for a Stepwise Approach

Interruption of anticoagulation temporarily increases thromboembolic risk, and continuing anticoagulation increases the risk of bleeding associated with invasive procedures. Both outcomes can increase mortality rates. We need a stepwise approach to help manage these patients.

Step 1: Estimate Bleeding Risk

Factors Contributing to Bleeding Risk

A higher bleeding risk confers a greater need for perioperative hemostasis and, hence, a longer period of anticoagulant interruption. Bleeding risk is dominated by the type and urgency of surgery; some patient comorbidities also contribute.

Bleeding risk is dominated by the type and urgency of surgery ([table 2](#)); some patient comorbidities (e.g., older age, decreased kidney function) and medications that affect hemostasis also contribute.

- **High risk:** High bleeding risk procedures include coronary artery bypass surgery, kidney biopsy, and any procedure lasting >45 minutes. In general, the anticoagulant must be discontinued if the surgical bleeding risk is high.
- **Low risk:** Low bleeding risk procedures include dental extractions, minor skin surgery, cholecystectomy, carpal tunnel repair, and abdominal hysterectomy. Individuals undergoing selected low bleeding risk surgery often can continue their anticoagulant.



Step 2: Estimate Thromboembolic Risk

Factors Contributing to Thromboembolic Risk

Those at very high or high thromboembolic risk should limit the period without anticoagulation to the shortest possible interval. If thromboembolic risk is transiently increased (e.g., recent stroke, recent pulmonary embolism), we prefer to delay surgery until the risk returns to baseline. It is importance to minimize the interval without anticoagulation.

- **Atrial fibrillation:** We estimate thromboembolic risk for patients with atrial fibrillation based on clinical variables, including age and comorbidities.
- **Prosthetic heart valve:** The risks of thromboembolism and perioperative management of patients with bioprosthetic and mechanical heart valves are discussed separately.
- **Recent thromboembolism:** The perioperative risk of venous thromboembolism (VTE) is greatest in individuals with an event within the prior three months, and those with a history of VTE associated with a high-risk inherited thrombophilia. Patients who require surgery within the first three months following an episode of VTE are likely to benefit from delaying elective surgery, even if the delay is only for a few weeks.

Step 3: Determine the Timing of Anticoagulant Interruption

Medication Considerations

HIGH BLEEDING RISK procedure			Day of surgery	No major bleeding		
Regular DOAC dose	X	X	X	X	Regular DOAC dose	Regular DOAC dose
LOW BLEEDING RISK procedure			Day of surgery	No major bleeding		
Regular DOAC dose	Regular DOAC dose	X	X	Regular DOAC dose	Regular DOAC dose	Regular DOAC dose

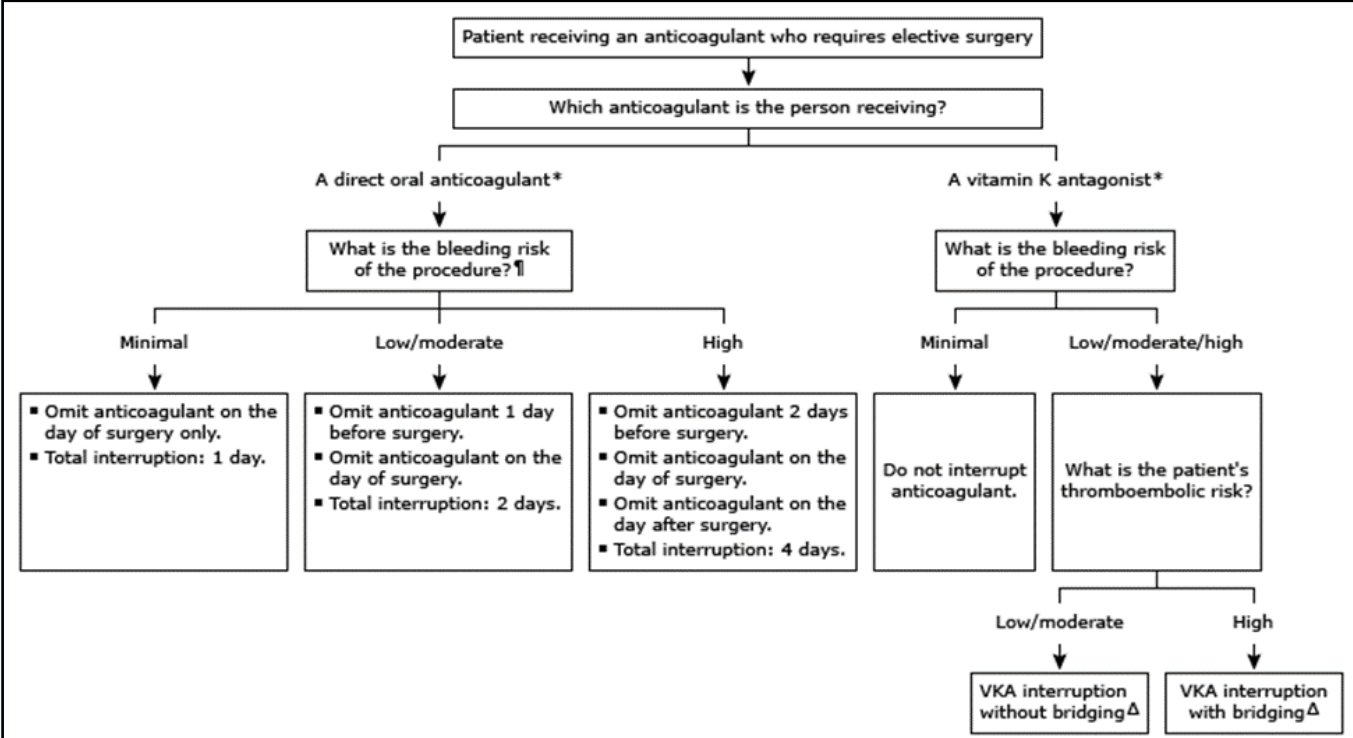
Note. From *Preoperative Medication Management* [Infographic], by Amjed Alnatsheh, 2021, Slideshare (<https://www.slideshare.net/amjedalnatsheh7/preoperative-medication-management>)

The timing of anticoagulant interruption depends on the specific agent the patient is receiving. [Warfarin](#) requires earlier discontinuation than the direct oral anticoagulants (DOACs; [apixaban](#), [dabigatran](#), [edoxaban](#), [rivaroxaban](#)). Additional considerations may be required in individuals with reduced kidney and/or liver function.

Step 4: Determine Whether to Use Bridging Anticoagulation

When is Bridging Beneficial?

For most patients, we do not use bridging anticoagulation because it increases bleeding risk without reducing the rate of thromboembolism. However, some patients on [warfarin](#) with an especially high thromboembolic risk (e.g., mechanical heart valve, recent stroke) may benefit from bridging.



DOAC: direct oral anticoagulant; VKA: vitamin K antagonist; CrCl: creatinine clearance; VTE: venous thromboembolism; IVC: inferior vena cava; LMW: low molecular weight.

* Anticoagulants

- Direct oral anticoagulants (DOACs) include dabigatran, apixaban, edoxaban, and rivaroxaban.
- Vitamin K antagonists include warfarin, acenocoumarol, phenprocoumon, and fluindione.

¶ The following applies to DOAC interruption

- These intervals are for individuals with normal kidney function and factor Xa inhibitors regardless of kidney function.
- For individuals with CrCl 30 to 50 mL/min receiving dabigatran, longer intervals are used (omit from 2 days before a low/moderate bleeding risk procedure; omit from 4 days before a high bleeding risk procedure).
- The perioperative management from the PAUSE study (a population with atrial fibrillation) can be applied to individuals who are receiving a DOAC for VTE that was >30 days prior. If the individual had a VTE within the prior 30 days, DOAC interruption should be individualized and may include placement of a temporary IVC filter or shorter periods of DOAC interruption.
- Bridging is not used for DOACs.

Δ The following applies to VKA interruption

- Discontinue warfarin 5 days before the procedure.
- If bridging is needed for a high thromboembolic risk patient, start LMW heparin at therapeutic dose approximately 3 days before surgery, with the last preoperative dose approximately 24 hours before surgery.
- Resume warfarin postoperatively once hemostasis is assured (typically the evening of the day of surgery or the day after surgery). Resume LMW heparin approximately 2 to 3 days after surgery (determined by the bleeding risk of the procedure) and discontinue LMW heparin after stable warfarin anticoagulation.
- The overlap period between LMW heparin and warfarin depends on the patient's thromboembolic risk.

Perioperative Management of Agents Affecting Hemostasis

Name of Class or Drug	Clinical Considerations	Recommended Strategy for Surgery with Brief NPO State	Recommended Strategy for Surgery w/ Prolonged NPO State
Aspirin	Continuation may cause perioperative hemorrhage. Discontinuation may increase the risk of vascular complications. Discussion with cardiologist appropriate for patients with cardiovascular indications.	Discontinue aspirin approximately seven days prior to noncardiovascular surgery.	Resume with oral intake.
P2Y12 receptor blockers (clopidogrel, prasugrel, ticlopidine, ticagrelor)	When used after cardiac stenting procedure, if discontinued can cause cardiac ischemia perioperatively. If continued can result in bleeding complications. Should discuss management with cardiologist.	Ideally, elective procedures should be delayed until the mandatory period of platelet inhibition with these agents is completed. When used for long-term stroke prophylaxis, should be discontinued 7 to 10 days. If discontinuing, stop clopidogrel and ticagrelor at least five days, prasugrel seven days, and ticlopidine 10 days before surgery. When restarting clopidogrel, consider using a loading dose.	Resume with oral intake.
Pentoxifylline	If being prescribed for alcoholic hepatitis, consult with prescribing hepatologist.	Take last dose the morning of surgery. However, there is generally no need to cancel/postpone surgery even if medication is continued due to low bleeding risk.	Resume with oral intake.

References

Alnatsheh, A. (2021). *Preoperative Medication Management* [Infographic]. Slideshare. <https://www.slideshare.net/amjedalnatsheh7/preoperative-medication-management>